

Anuj Sethia

Quantum Engineer

anujsethia66@gmail.com | anujsethia.github.io | linkedin.com/in/anuj-sethia-713218142/ | +1 403 399 9296

HIGHLIGHTS

Competences: Quantum Optics, Integrated Photonics, Nano-Fabrication, Cryogenics, RF Electronics

Skills: Phyton, QuTip, Pennylane, Qiskit, Matlab, Lumerical, Comsol, IPKISS, Virtuoso, Solidworks

EDUCATION

Ph.D. in Physics (Fastracked from M.Sc.)

Aug 2021 - Sep 2026

University of Calgary, Calgary, Canada

Awards: Alberta Innovates Graduate Student Scholarship, Alberta Graduate Excellence Scholarship, Faculty of Graduate Studies Scholarship, MicroFAB Access by CMC Microsystems

B.Tech. in Mechanical Engineering with Minor in Physics

Jul 2016 - Aug 2020

Indian Institute of Technology Bombay, Mumbai, India

Awards: Undergraduate Research Award

KEY PROJECTS

On-chip Quantum Memory

Jun 2024 - Aug 2026

Advisor: Prof. Daniel Oblak | University of Calgary

- Established cryogenic experimental capabilities for low loss coupling of photonic chips
- Characterize Erbium-doped Lithium Niobate waveguides to optimize coherence for quantum memory
- Maximize the memory storage efficiency by implementing on-chip resonant cavity enhancement

Nano-Fabrication

Jun 2024 - Aug 2026

Advisor: Prof. Daniel Oblak | University of Calgary

- Pioneered multi-step nano-fabrication workflows for Thin-Film Lithium Niobate photonics devices
- Engineered Electron-Beam Lithography (EBL) and Reactive-Ion Etching (RIE) recipes, eliminating stitching errors and achieving nanometer-scale sidewall smoothness.
- Achieved propagation losses of 0.25 dB/cm, enhancing the efficiency of on-chip quantum systems.

Millisecond Storage Quantum Memory

Jan 2025 - Dec 2025

Advisor: Prof. Daniel Oblak | University of Calgary

- Engineered an alignment-free interface at cryogenic temperature for scalable quantum repeaters
- Pioneered a telecom quantum memory utilizing Er ions, with longest reported storage time of 1 ms
- Achieved a 60-mode storage capacity through spectral-temporal multiplexing, addressing the critical requirement for scalable quantum network architectures.

Silicon Photonics

Jan 2023 - Jun 2024

Advisor: Prof. Daniel Oblak | University of Calgary

- Modeled and optimized the performance of key Silicon Photonics devices using Lumerical
- Developed Python-based die testing platform for streamlined data acquisition and rapid prototyping
- Demonstrated generation of on-chip entangled photon pairs for quantum communication protocols

Measurement Device Independent Quantum Key Distribution

Sep 2021 - Jun 2023

Advisor: Prof. Daniel Oblak | University of Calgary

- Integrated high-bandwidth electro-optic devices for high-fidelity and fast time-bin qubit generation
- Collaborated with a diverse team of 6 engineers to develop a minimum viable product for QKD
- Pioneered and patented innovative control feedback mechanisms, enabling stable and reliable long-distance quantum communication over fiber networks

Quantum Approximate Optimization Algorithm

Sep 2022 - Dec 2022

Advisor: Prof. Barry Sanders | University of Calgary

- Studied Variational Quantum Algorithms such as quantum annealing and QAOA for Noisy Intermediate Scalable Quantum devices to solve combinatorial optimization problems
- Researched challenges of implementing QAOA on Ion-trap & Superconducting quantum computers

Optimal Quantum Control

Jan 2020 - May 2020

Advisor: Prof. Sai Vinjinampathy | IIT Bombay

- Studied time-optimal unitary transformation, crucial for error mitigation in quantum computers
- Explored the Quantum Brachistochrone equation for its relevance in Quantum Control problems
- Analysed Quantum Speed Limit and the corresponding time lower bound for unitary transformations

PROFESSIONAL EXPERIENCES

Research Assistant | QNu Labs, Bangalore

Jun 2020 - Jul 2021

Quantum Key Distribution:

- Conceived and built a modeling framework using Matlab to simulate real-world QKD systems
- Analyzed information-theoretic security of DPS-QKD protocol from advanced eavesdropping attacks
- Extended the range of key exchange beyond 100 km via systematic optimization of parameters
- Initiated conceptualization and early development of a quantum-secure video conferencing service

Quantum Random Number Generator:

- Spearheaded product development of a QRNG in collaboration with a diverse cross-functional team
- Enhanced the entropy extraction scheme for a ten-folds increase in post-processed throughput rate
- Performed comprehensive benchmarking and statistical randomness tests to meticulously assess both the system's performance and the quality of output randomness

KEY PROFESSIONAL DEVELOPMENT

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| • Reviewer @ Physical Review Letters, American Physical Society | Jul 2024 - Present |
| • Awarded Evolve to Innovate for translating research into impact | Jan 2026 - May 2026 |
| • Member @ Research Innovation & Scholarship Committee, UCalgary | Sep 2025 - Apr 2026 |
| • Member @ Quantum Horizons Alberta Director Hiring Committee, UCalgary | Sep 2025 - Apr 2026 |
| • Participant @ GenQ Quantum Hackathon | Oct 2024 |
| • Finalist @ Three Minute Thesis Competition | Apr 2024 |
| • Best Poster Presentation @ IEEE Photonics Conference, Orlando | Nov 2023 |
| • Best PIC Design Award @ Quantum Photonics Workshop, UBC, Vancouver | Jan 2023 |
| • Secretary & Treasurer @ Calgary Optics and Photonics Students Society | Jan 2022 - Jan 2023 |
| • Mentor @ Student Mentorship Program, IIT Bombay | Aug 2019 - Jul 2020 |

PUBLICATIONS AND PATENTS

1. **Sethia, A.**, Kamel, N. G., & Oblak, D. (2025). On-Demand millisecond storage of Spectro-Temporal multimode telecom photons. [arXiv preprint]
2. Singh, A., **Sethia, A.**, Esmaeilifar, L., Valivarthi, R., Sinclair, N., Spiropulu, M., & Oblak, D. (2025). Photonic quantum information with time-bins: Principles and applications. [arXiv preprint]
3. Smith, J., Oblak, D., **Sethia, A.**, Chenini, H., Singh, A. (2023) Apparatus and Methods for Ongoing Feedback during Quantum Key Distribution. US Patent 63/527,960, [Pending]
4. **Sethia, A.**, & Banerjee, A. (2022). A MATLAB based modeling and simulation package for DPS-QKD. Journal of Modern Optics, 69(7), 392–402
5. Banerjee, A., **Sethia, A.**, Mogiligidha, V., Krishnan, R. K., AR, M., Rajamani, S., & Shenoy, V. (2021). Experimental demonstration of high-entropy time of arrival based optical QRNG qualifying stringent statistical tests [arXiv preprint]

